

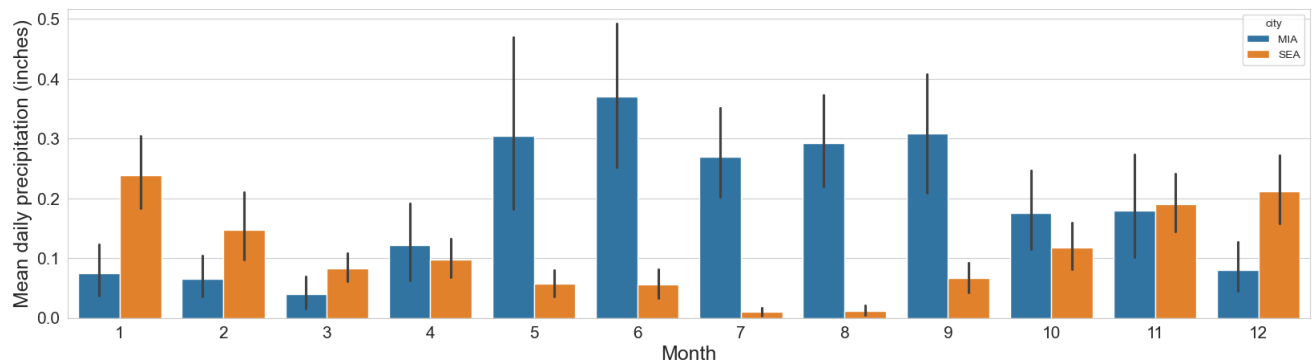
Seattle Weather Project

Seattle has a reputation for being one of the rainiest cities in the United States. A lot of people picture gray skies and drizzle for most of the year. Miami, on the other hand, is known for sunshine and beaches, so most people would probably guess that it rains less there. In this project I use daily weather data from 2018 to 2022 from NOAA SeaTac Airport and Miami Airport to check whether it actually rains more in Seattle than in Miami. I look at the question in two ways. First, I compare the amount of precipitation in each city. Second, I compare how often it rains, meaning the proportion of days that had any precipitation. These two things do not have to agree, since a city could have a lot of light rainy days or a smaller number of very heavy ones.

For the analyses of my data, I compared the cities using summary statistics, line plots, bar graphs, and box plots, both overall and by month/season. I created an any_precipitation variable to compare how often it rains in each city, so I can observe the frequency of rain. To make sure my comparisons were statistically significant, I used independent samples t-tests to compare mean daily precipitation between the cities in each month and proportions z-tests to compare the proportion of days with precipitation between the cities in each month.

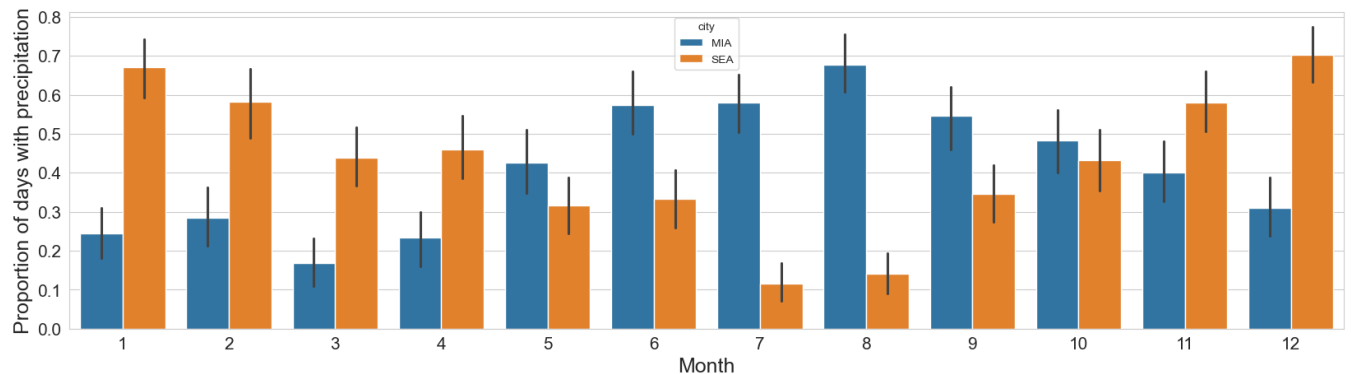
My results were as follows:

Amount: Miami averages more precipitation per day than Seattle (0.19 inches vs. 0.11 inches). Seattle has a significantly higher mean in January, February, March, and December, while Miami has a significantly higher mean from May through September.



Frequency: The two cities have precipitation on about the same proportion of days overall

(42.6% in Seattle and 41.1% in Miami). Seattle has rain significantly more often from November through April, and Miami has rain significantly more often from May through September.



In conclusion, it does not rain more in Seattle than in Miami. Miami gets more precipitation on average (0.19 inches per day compared to 0.11 inches per day in Seattle), and the two cities have precipitation on about the same proportion of days (about 42% in Seattle and 41% in Miami). However, the answer changes depending on the season. In the winter months, Seattle gets significantly more rain than Miami and has rain on many more days. In the summer it is the opposite, and Miami gets much more rain and has rain more often. Miami's summer storms are also much heavier, which is why Miami's total is so much higher. This may help explain Seattle's reputation. Seattle has rain on about two out of every three days in December and January, so it feels like it rains all the time during the winter, even though it gets less rain than Miami over the whole year. One limitation of this analysis is that each city only uses one weather station at the airport, which may not match the weather in every part of the city. The data also only covers five years, so a longer period would give a more reliable answer.